Structural Hawkes Ambiguity in Near-Critical Market Making

Abstract

This paper studies a narrow source of model risk in market making: uncertainty about the self-excitation structure of Hawkes order flow. The structure is encoded by a branching matrix Γ, while the distance to instability is measured by the Perron root $\rho(\Gamma)$. In a reduced robust inventory model, ambiguity in Γ is amplified through the Hawkes resolvent $(I-\Gamma)^{-1}$, so near-critical calibration error can become an economic risk premium. The critical exponent depends on normalization: relative critical-slack ambiguity gives a square-law reduced premium, while absolute branching-matrix ambiguity contributes one additional derivative factor when $\epsilon = o(1-\rho)$.

The paper gives a reduced-form spectral theorem and conditional robust-control bridges under stated compact-PDMP, Lyapunov, weighted-comparison, and smooth-interior assumptions. It then evaluates robust and nominal quoting policies under common random numbers, Ogata event times, public LOBSTER sample replays, Binance aggregate trades, and crypto order-book samples. The empirical claim is deliberately limited: structural Hawkes ambiguity is visible in these near-critical stress tests, but public Hawkes calibrations remain diagnostic inputs rather than complete execution models.

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Reproducibility package. Code, tests, TeX sources, derived tables, and derived figures: `github.com/mayank172900/mesa-sifin-reproducibility`.

1 Introduction

When liquidity disappears during stress, the usual explanations are volatility, inventory pressure, or lower fill probabilities. This paper focuses on a smaller but sharper mechanism. If order flow is self-exciting and already close to criticality, then uncertainty about the excitation structure can become expensive by itself. The same estimation error in Γ may be harmless far from criticality and serious when $\rho(\Gamma)$ is close to one.

The contribution is narrow by design. Hawkes market making, robust market making, near-critical Hawkes limits, and learned limit-order-book simulators are all active areas. This paper studies their intersection: structural Hawkes ambiguity in market making, the spectral amplification law it creates, and reproducible diagnostics showing why near-critical Hawkes calibrations should be stress-tested before being used inside a trading simulator.

The paper makes four contributions. It identifies the resolvent channel through which branching-matrix uncertainty amplifies risk-premium proxies. It separates relative critical-slack ambiguity from absolute branching-matrix ambiguity, which changes the critical exponent. It develops the reduced control problem through a sequence of conditional finite-state, compact-PDMP, Lyapunov, weighted-comparison, and interior-optimizer results. Finally, it reports a reproducible experimental path from synthetic scaling tests to robust Bellman policies, event-time execution stress tests, public LOBSTER replays, Binance aggregate-trade diagnostics, and crypto depth sanity checks.

Reader's map

The main idea can be understood before reading the mathematical appendix that follows the paper in the combined SSRN PDF. If order flow is described by a Hawkes branching matrix Γ, then the long-run effect of arrivals is filtered through the resolvent $(I-\Gamma)^{-1}$. As $\rho(\Gamma)$ moves toward one, this resolvent grows in the leading Perron direction. For a market maker uncertain about the excitation matrix, this makes near-critical calibration error more than a small nuisance parameter. It can be magnified into a meaningful economic risk.

The rest of the paper proceeds in three layers. The first is spectral and gives the normalization-dependent amplification laws. The second is a robust-control bridge showing how the spectral effect enters reduced Bellman/HJBI equations for market making. The third is empirical and computational. The experiments do not form a production execution engine. They ask whether the same risk layer remains visible under finite-state policies, event-time simulation, public LOBSTER replay, crypto clustering diagnostics, calibration noise, and ablation tests.

Main claim. MESA–market-making excitation-structure ambiguity–does not claim that Hawkes models fully describe limit-order-book execution. It says that, conditional on using a Hawkes structural layer, ambiguity in Γ is amplified near criticality, and the amplification exponent depends on how that ambiguity is normalized. The theory establishes this reduced-form spectral claim under stated assumptions; the experiments test whether the mechanism is visible in the stress designs reported below.

2 Related Work and Positioning

Nearby literatures approach this problem from several directions. Point-process and Hawkes market-making models study weakly consistent LOB control [2], Hawkes impulse control [10], and adversarial or reinforcement-learning variants [6]. A separate LOB-simulation and forecasting literature uses neural Hawkes, multivariate Hawkes, state-dependent Hawkes, and non-Hawkes realism-focused simulators to improve realism and calibration [7, 3, 4, 11, 12, 13]. Recent work on robust quote refusal and latency-aware policies addresses different forms of execution risk [5, 9]. Empirical Hawkes analyses also motivate concern about near-critical financial event flow [16], while nearly unstable Hawkes limits clarify what happens close to criticality [14, 15]. MESA is narrower than these programs. Throughout the paper, MESA means market-making excitation-structure ambiguity: a spectral ambiguity layer for Hawkes branching matrices. It does not propose a new full LOB simulator or a competing RL agent; it isolates how uncertainty in the branching matrix is amplified when the Hawkes system is near instability.

The reproducibility package records the source URLs, visible code links, and implementation notes used to construct the comparison. The intended position is therefore modest: MESA is a spectral ambiguity-risk layer that can be attached to richer simulators, not a substitute for them.

Work	Hawkes MM	Robust MM	Γ uncertainty	Near-critical scaling	MESA position
Law–Viens 2019/2020	no	no	no	no	Marked point-process/HJB-QVI LOB MM benchmark; MESA adds Hawkes branching-matrix ambiguity.
Wang–Ventre–Polukarov 2023/2025 quote/no-quote	no	yes	no	no	Robust quote-refusal/single-sided quoting benchmark; Hawkes branching uncertainty is outside its scope.
Wang–Ventre–Polukarov 2024/2025 Hawkes ARL	yes	yes	no	no	A close Hawkes/ARL policy benchmark; no branching-matrix ambiguity amplification law.
Jain et al. 2025 Hawkes impulse control	yes	no	no	no	A close HJB-QVI benchmark; robustness to Γ is not the main object.
Lalor–Swishchuk 2025 neural Hawkes LOB	yes	no	no	no	Neural Hawkes LOB simulator/DRL market-making baseline; ambiguity amplification is complementary.
Kumar 2021/2024 Deep Hawkes	yes	no	no	no	Deep Hawkes high-frequency MM/simulation benchmark; MESA adds structural Γ ambiguity diagnostics.
Guo–Lin–Huang 2023 Attn-LOB DRL [8]	no	no	no	no	LOB-feature DRL baseline; focus is alpha learning rather than Hawkes structure.
Jiang et al. 2025 latency-aware RL	no	no	no	no	Latency/inventory-aware neural MM benchmark; robust Hawkes ambiguity is not modeled.
Raffaelli et al. 2026 multivariate Hawkes BTC LOB	no	no	no	no	Recent multivariate Hawkes BTC LOB calibration/forecasting study; robust control is outside its scope.
El Karmi 2025 deterministic Hawkes LOB simulator	no	no	no	no	Simulator-realism baseline with Hawkes-driven order flow; robustness is complementary.
Noble–Rosenbaum–Souilmi 2026 realism gap	no	no	no	no	Recent simulator-realism and execution-sensitivity study; MESA is a complementary risk layer.
Kimura 2026 state-dependent Hawkes	no	no	no	no	State-dependent Hawkes LOB with local supercriticality evidence; MESA targets robust Γ ambiguity.
Szymanski–Xu 2025 nearly unstable Hawkes	no	no	no	yes	Near-critical Hawkes scaling limits; MESA adds market-making control and Γ-ambiguity premiums.
El Karmi 2026 bivariate nearly unstable Hawkes	no	no	no	yes	Recent bivariate near-critical Hawkes limit theory; MESA targets robust market-making premiums.
MESA reduced-form package	yes	yes	yes	yes	Reduced-form theorem, policy stress tests, and public-data calibration diagnostics.

Table 1: Comparison with nearby work. The point is not to claim simulator primacy, but to isolate the structural Hawkes ambiguity layer.

3 Model

Let $N_t^{k,s}$ denote marked order arrivals by type k and side s. The exponential Hawkes intensity is

$$\lambda_t^{k,s} = \mu^{k,s} + \sum_{j,\sigma} \int_0^{t^-} \alpha_{j,\sigma}^{k,s} \beta e^{-\beta(t-u)}\, dN_u^{j,\sigma}.$$

Thus Γ records the integrated excitation from each event type into each other event type. Stability is governed by its Perron root: when $\rho(\Gamma) < 1$, the stationary mean intensity is

$$\bar{\lambda} = (I - \Gamma)^{-1}\mu.$$

This resolvent is the channel through which small errors in Γ become large near criticality.

A market maker chooses bid and ask offsets $u_t = (\delta_t^b, \delta_t^a) \in \mathcal{U}$ with inventory $q_t \in \{-Q, \ldots, Q\}$. Nature chooses a structural model from an ambiguity set

$$\mathcal{A}_\epsilon = \{\Gamma \geq 0 : \|\Gamma - \widehat{\Gamma}\|_F \leq \epsilon,\ \rho(\Gamma) \leq \rho_{\max} < 1\}.$$

The asymptotic statements are interpreted along Perron-regular local sequences with $\rho_{\max}$ allowed to approach one, while numerical experiments use finite stable stress scenarios below one. For absolute ambiguity, the derivative law is local: if ϵ is not smaller than the remaining critical slack, the perturbation can saturate at the stability cap and the simple $(1-\rho)^{-3}$ expansion no longer applies. For numerical work, this ambiguity set is discretized into the nominal matrix and a small collection of Perron-aligned boundary perturbations.

4 Theory and Main Results

Assumption 1 (Perron regularity)**.** *Γ is nonnegative, irreducible, and primitive, with Perron root $\rho < 1$. Let $\operatorname{gap}(\Gamma) = \rho - \max_{j>1} |\lambda_j(\Gamma)|$. Assume $\operatorname{gap}(\Gamma) \geq g_0 > 0$, the Perron root is simple, the Perron eigenprojection $r\ell^\top$ is uniformly bounded under the normalization $\ell^\top r = 1$, and the reduced resolvent on the non-Perron spectral subspace is uniformly bounded. This rules out non-Perron conditioning from carrying the same singular order as the Perron pole.*

Proposition 1 (Resolvent amplification)**.** *Under Perron regularity,*

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho(\Gamma)} + O(1/\text{gap}).$$

Consequently, price-risk loadings whose Perron projection is bounded away from zero on the Perron-regular class are exposed to the same leading near-critical blow-up.

Proposition 2 (Perturbation direction)**.** *For a small perturbation E,*

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2/\text{gap}).$$

The first-order Frobenius direction that maximally increases the Perron root is $E^\star = \ell r^\top / \|\ell r^\top\|_F$, distinct from the resolvent projection $r\ell^\top$.

Theorem 1 (Reduced-form normalization-dependent ambiguity premium)**.** *Suppose the robust half-spread correction is locally linear in an effective Hawkes variance proxy proportional to $(1-\rho)^{-2}$. The square-law exponent uses this normalized-throughput/resolvent-squared risk proxy; alternative* *raw count-variance conventions may change the baseline exponent, but not the distinction between relative-slack and absolute-*Γ *ambiguity. Then relative critical-slack ambiguity,* $\Delta\rho = O(\epsilon_{\mathrm{rel}}(1-\rho))$*, gives*

$$\psi = O\left(\frac{\epsilon_{\mathrm{rel}}}{(1-\rho)^2}\right).$$

Absolute branching-matrix ambiguity, $\Delta\rho = O(\epsilon)$ *with* $\epsilon = o(1-\rho)$*, gives the more singular derivative law*

$$\psi = O\left(\frac{\epsilon}{(1-\rho)^3}\right).$$

Matching lower bounds require Perron-visible loadings, a feasible Perron-aligned perturbation, an active worst-case selector, and an interior quote regime before spread caps or no-quote constraints bind.

Remark 1. *The theorem is stated at the reduced-form level because the amplification mechanism is spectral, while the market-making problem can be embedded in several control formulations. The mathematical appendix gives scalar and finite-dimensional Perron-visible versions of the bridge, a finite-state Bellman verification theorem, and a compact continuous-intensity HJBI verification theorem for a truncated multitype Hawkes control game. It also records the Lyapunov, weighted-comparison, and local differentiability arguments used to connect the reduced premium calculation to smooth interior quote optimizers. The main limitations are active-set changes from no-quote or spread caps, the compact-PDMP comparison hypotheses, and extensions to state-dependent, power-law, or learned Hawkes kernels.*

5 Computational Design

The empirical design combines complementary simulations, control solvers, and replay diagnostics. Its roles are to test the predicted critical scaling, evaluate robust quote policies under controlled Hawkes variation, and compare the resulting behavior with public limit-order-book data. The components are:

1. Scaling diagnostics for scalar and multitype resolvent proxies.
2. Common-random-number policy evaluation under full simulated Hawkes paths.
3. A finite-scenario robust inventory Bellman solver for a scalar reduced state with side-level quote/no-quote actions.
4. A reduced event-queue backtest on Ogata event times with queue-ahead mechanics and no-quote side-time accounting.
5. Displayed-depth, residual-volume priority-aware level-10, and deepest-public level-50 LOBSTER replays on public message/order-book streams.
6. Observable level-10 order-book reconstruction from messages, checked against official LOBSTER snapshots.
7. Ogata-thinning validation and time-step bias diagnostics.
8. LOBSTER Hawkes diagnostics over event type, time scale, marks, and side labels.
9. Binance aggregate-trade event diagnostics for public crypto clustering.

The Bellman recursion uses side-specific quote/no-quote controls: on each side the market maker either posts at a chosen offset or withdraws liquidity.

$$h_n(q) = \max_{a^b,\delta^b,a^a,\delta^a} \min_{\Gamma\in S} \Big\{ -\Delta t \left(\phi q^2 + \frac{\gamma}{2}\sigma^2(\Gamma) q^2\right) + p_b(\delta^b + h_{n+1}(q+1)) + p_a(\delta^a + h_{n+1}(q-1)) + (1-p_b-p_a)h_{n+1}(q)\Big\}.$$

The indicators $a^b, a^a \in \{0,1\}$ encode side activity; when a side is inactive, its fill probability is set to zero.

Theorem 2 (Finite-state Bellman verification)**.** *Consider a finite state space encoding inventory and Hawkes regimes, a nonempty finite quote/no-quote action set at each state, a conservative finite-state generator $L^{u,\Gamma}$, bounded rewards, bounded terminal payoff, and a finite rectangular ambiguity set S of stable branching matrices. The finite-regime robust control problem has the max–min value. In continuous time its value is the unique bounded solution of the finite-state HJBI ODE*

$$-\partial_t V(t,x) = \max_{u\in U(x)} \min_{\Gamma\in S} \left\{ r(x,u,\Gamma) + \sum_y L^{u,\Gamma}(x,y)V(t,y) \right\}, \qquad V(T,x) = g(x),$$

On a time grid, the analogous backward Bellman recursion is exact for the chosen stochastic one-step kernel, and for an explicit Euler kernel requires Δt small enough that $I + \Delta t L^{u,\Gamma}$ is stochastic. Deterministic Markov selectors attain the finite max–min. This identifies the finite-scenario quote/no-quote dynamic program with a discretized robust finite-state game. The continuous-intensity Hawkes HJBI is not covered by this finite-state argument; the compactness, Lyapunov, and weighted-comparison results below supply the bridge to that setting.

Theorem 3 (Conditional compact continuous-intensity HJBI verification)**.** *Consider the truncated multitype Hawkes control state $X_t = (q_t, \lambda_t)$, with finite inventory $q_t \in \{-Q, \dots, Q\}$ and compact intensity domain $\lambda_t \in K \subset \mathbb{R}^d_+$. Between jumps, $d\lambda_t = \beta(\mu - \lambda_t)\,dt$, and a mark-$i$ jump maps λ to $\Pi_K(\lambda + \beta\Gamma e_i)$, where Π_K is a fixed Lipschitz truncation map into K. Suppose the quote action set and the rectangular ambiguity set $\mathcal{A}$ are compact, all rewards, terminal payoffs, and controlled mark intensities are bounded and continuous, the rates and jump maps are uniformly Lipschitz on each inventory slice, $\sup_{\Gamma\in\mathcal{A}} \rho(\Gamma) < 1$, and the compact PDMP game satisfies the standard dynamic-programming and bounded-comparison principles for the rectangular admissible class, using standard PDMP, stochastic-control, and viscosity-solution machinery [20, 19, 18]. Then the robust market-making game has a bounded uniformly continuous value function, and it is the unique bounded viscosity solution of*

$$-\partial_t V = \sup_u \inf_{\Gamma\in\mathcal{A}} \left\{ r(q,\lambda,u,\Gamma) + \mathcal{L}^{u,\Gamma} V(q,\lambda) \right\}, \qquad V(T,q,\lambda) = g(q,\lambda),$$

where $\mathcal{L}^{u,\Gamma}$ is the Hawkes PDMP generator. If the value is classically differentiable and the Hamiltonian has measurable saddle selectors, those selectors verify the robust value. Consequently, the truncated continuous-intensity game admits the claimed HJBI characterization under the stated compact-PDMP hypotheses. Passing to the global, untruncated near-critical model still requires the stability and comparison arguments developed next.

Theorem 4 (Conditional untruncated Hawkes Lyapunov bridge)**.** *For the unprojected Hawkes intensity state $\lambda_t \in \mathbb{R}^d_+$, suppose there exist $v > 0$ and $\kappa > 0$ such that $\Gamma^\top v \le (1-\kappa)v$ for* *every $\Gamma \in \mathcal{A}$, fix $p > 1$, and suppose controlled mark rates satisfy $0 \le \Lambda_i(q, \lambda, u, \Gamma) \le \lambda_i$. Then $W_v(\lambda) = 1 + v^\top \lambda$ satisfies a uniform Foster–Lyapunov drift,*

$$\mathcal{L}^{u,\Gamma} W_v(q, \lambda) \le C_0 - C_1 W_v(\lambda),$$

and, after changing constants, the standard polynomial-weight version

$$\mathcal{L}^{u,\Gamma} W_v(\lambda)^p \le C_{0,p} - C_{1,p} W_v(\lambda)^p.$$

Hence the untruncated controlled Hawkes PDMP is nonexplosive and has finite pth weighted maximal moments on finite horizons. For rewards and terminal payoffs with at most linear W_v-growth, compact truncations $\{W_v \le R\}$ have exit probability at most $C_{T,p} W_v(\lambda)^p / R^p$, and the payoff tail error is of order R^{1-p}. Thus the compact projection is not needed for stability or tightness. What remains is uniqueness of the limiting HJBI on the unbounded intensity domain, addressed by the weighted comparison result below.

Theorem 5 (Sufficient weighted-comparison bridge for the untruncated HJBI)**.** *Following the standard weighted-barrier and doubling-of-variables logic used in viscosity comparison arguments [18, 19], assume common Lyapunov subcriticality, compact ambiguity, and either a fixed compact control set or a compact control correspondence continuous in Hausdorff distance, continuous locally Lipschitz controlled rates with $0 \le \Lambda_i \le \lambda_i$, Lipschitz affine Hawkes jump maps, linearly growing rewards, and the weighted boundary condition*

$$\limsup_{R \to \infty} \sup_{\substack{t \in [0,T],\, q \in \{-Q,\dots,Q\} \\ W_v(\lambda) \ge R}} \frac{U(t, q, \lambda) - V(t, q, \lambda)}{W_v(\lambda)} \le 0$$

for every subsolution U and supersolution V being compared. Then any upper-semicontinuous weighted-growth subsolution of the untruncated HJBI is dominated by any lower-semicontinuous weighted-growth supersolution with larger terminal data. Hence, within any W_v-weighted subclass with a fixed boundary condition at infinity, the untruncated robust Hawkes control value is the unique viscosity solution, and compact truncation solutions converge to it locally uniformly.

Theorem 6 (Interior robust quote optimizer sensitivity)**.** *Fix the state and adjoint variables in the HJBI Hamiltonian, and let θ parameterize either the branching matrix or the center of the ambiguity set. Suppose that, near (δ_0, θ_0), the robust reduced Hamiltonian $\mathfrak{H}(\delta, \theta) = \inf_{\Gamma' \in \mathcal{A}(\theta)} \mathcal{H}(\delta, \Gamma')$ has a unique C^1 active worst-case selector, the quote constraints are inactive, $\mathfrak{H}$ is C^2 in δ with continuous mixed derivative $D^2_{\delta\theta}\mathfrak{H}$, δ_0 is the unique local maximizer at θ_0, and $D^2_{\delta\delta}\mathfrak{H} \preceq -mI$ locally. Then, for θ near θ_0, there is a unique interior quote maximizer $\delta^*(\theta)$ in the corresponding quote neighborhood, and the map $\theta \mapsto \delta^*(\theta)$ is locally C^1 with*

$$D_\theta \delta^*(\theta_0) = -\left[D^2_{\delta\delta}\mathfrak{H}(\delta_0, \theta_0) \right]^{-1} D^2_{\delta\theta}\mathfrak{H}(\delta_0, \theta_0).$$

In the one-dimensional half-spread case, if e is a Perron-adverse structural direction and the directional mixed derivative in direction e is nonnegative, then the robust interior half-spread widens to first order. No-quote boundaries, spread caps, and changes in the active adversary fall outside this smooth interior calculation and instead require active-set analysis.

The computations use vectorized quote-grid evaluation and scenario-level parallelism. To make policy comparisons attributable to the policy rather than simulation noise, paired evaluations use fixed seeds and common random numbers.

6 Experiments

We first test the critical-scaling predictions in controlled ablations, then evaluate the robust quote policies in simulated and replayed order-flow environments. The experiments are ordered from mechanism-level checks to market-data diagnostics.

6.1 Scaling Ablations

Object	Family	Mean slope
Relative-slack premium	scalar	-2.000
Absolute-Gamma derivative	scalar	-3.000
Matrix resolvent	rank1	-3.191
Matrix resolvent	block	-3.213
Matrix resolvent	near-degenerate	-3.202
Matrix resolvent	sparse	-3.283

Table 2: Critical exponent ablation. Relative critical-slack ambiguity and absolute branching-matrix ambiguity produce the distinct blow-up rates predicted by the reduced-form theory.

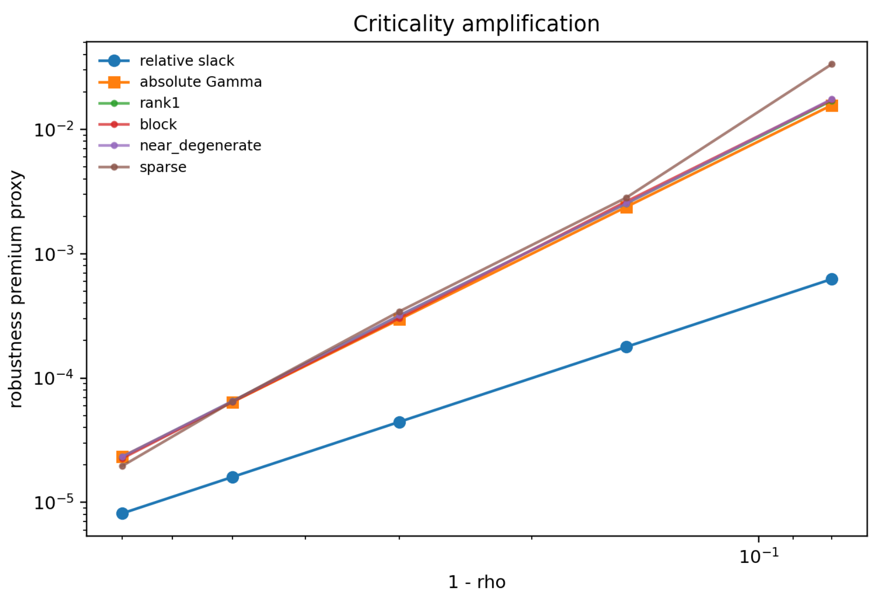

Figure 1: Criticality amplification under relative-slack ambiguity, absolute-Γ perturbations, and matrix-resolvent stress designs.

To separate the exponent from the particular stress design, we use a targeted spectral-gap and visibility ablation based on a two-mode branching matrix with eigenvalues ρ and $1 - k(1 - \rho)$ for $k \in \{2, 4, 8\}$. When the payoff loading is Perron-visible and the ambiguity set is aligned with the Perron direction, the estimated slope is -2.00, with median normalized premium ratio 2.8125. A weakly Perron-visible loading recovers the same exponent after visibility normalization. By contrast, a Perron orthogonal loading has no positive Perron-aligned premium. When the adversary is instead allowed to move the second eigenmode, the orthogonal loading again has slope -2.00 as that mode approaches criticality. For weakly visible loadings, however, the median normalized second-mode coefficient falls from 2.639 at $k = 2$ to 0.0365 at $k = 8$. The lower-bound statement therefore requires both Perron visibility and an ambiguity set that activates the Perron direction; otherwise the leading coefficient can vanish or be transferred to another near-critical mode.

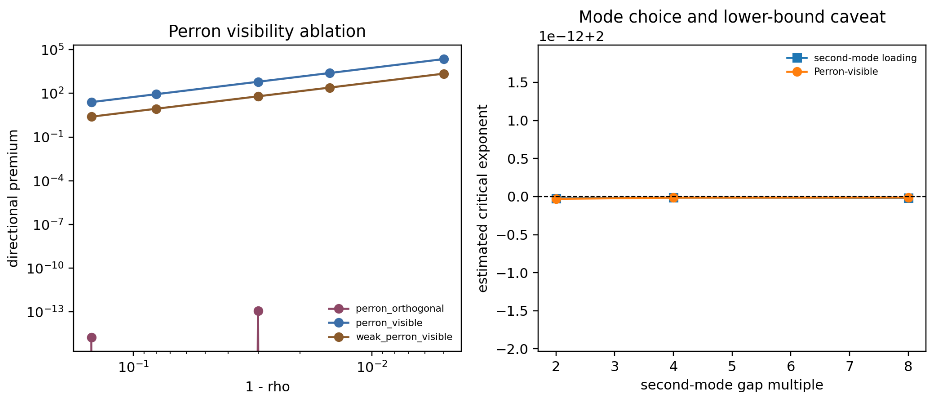

Figure 2: Spectral-gap and Perron-visibility ablation. Perron-visible loadings recover the square law under aligned ambiguity, while orthogonal loadings only respond to second-mode perturbations.

6.2 Policy Ablations

Policy	Certainty equivalent	CVaR5	Mean diff vs nominal	Paired 95% CI
robust_gamma	-13.944	-38.073	0.012	[0.005, 0.020]
robust_vol_only	-13.956	-38.087	0.000	[0.000, 0.000]
nominal_hawkes	-13.956	-38.087	baseline	–
robust_gamma_abs	-13.624	-37.321	0.306	[0.257, 0.359]
known_true_gamma_no_ambiguity	-13.952	-38.082	0.004	[-0.002, 0.011]
as_poisson	-13.974	-38.113	-0.017	[-0.026, -0.008]
liquidity_guard	-13.624	-37.321	0.306	[0.256, 0.355]

Table 3: Common-random-number policy ablation.

The intervals are paired bootstrap intervals over common-random-number paths, so each policy is compared with nominal on the same simulated realizations. In the main ablation, the absolute-Gamma and liquidity-guard policies have the highest reported certainty equivalents, CVaR5 values, and paired mean-wealth differences. The relative-slack robust-Gamma policy is closer to nominal, with a smaller but positive paired mean-wealth difference. We do not interpret the known-true-Γ benchmark as an oracle optimum: it is given the stressed ρ, but it does not carry an ambiguity premium, so it under-hedges model risk in this stress design. The table is best read as evidence that the policy response depends materially on the ambiguity normalization and on the withdrawal rule.

6.3 Policy Time-Step Audit

The simulator validation below shows that near-critical Hawkes counts can be sensitive to bin width, so we repeat the focused policy comparison at $\Delta t \in \{0.04, 0.02, 0.01\}$ for the stressed $\epsilon = 0.02$, $\hat{\rho} = 0.92$ scenario. Across the tested time steps, the relative-slack robust-Gamma policy keeps a positive mean-wealth advantage over nominal Hawkes: 0.042–0.060. The absolute-Gamma policy produces much larger mean-wealth gains, 1.154–1.366, but with negative lower paired 5% quantiles, consistent with the broader point that ambiguity normalization changes the economic policy, not only the scale of the premium.

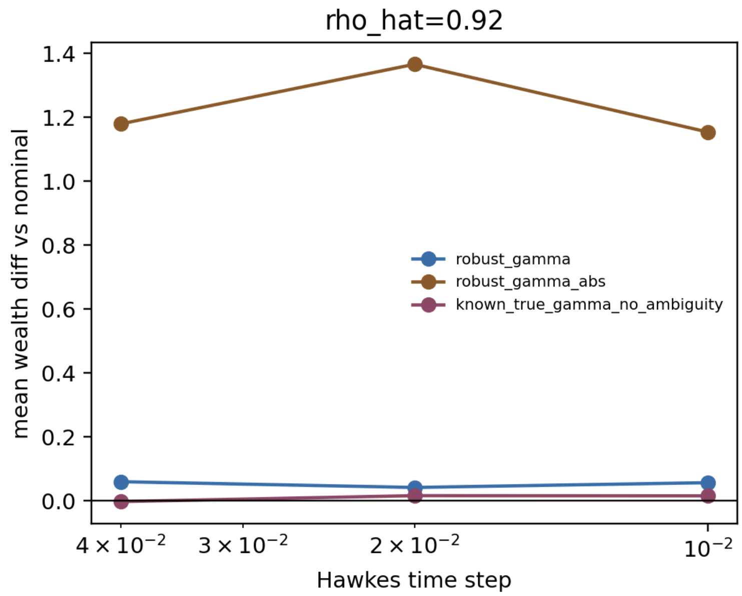

Figure 3: Focused policy time-step check. Positive values mean the policy outperforms nominal Hawkes on the same scenario.

6.4 Ogata Arrival Audit

We next compare the fast binned Hawkes recursion with event times generated by Ogata thinning and then binned onto the same decision grid. In the $\epsilon = 0.02$, $\hat{\rho} = 0.92$ check, the discrete recursion produces more events than the Ogata reference: mean total events are 226.8 under the discrete recursion and 185.5 under Ogata-binned arrivals. The relative-slack robust-Gamma policy is positive relative to nominal under discrete arrivals, with mean-wealth advantage 0.030, but is essentially flat and slightly negative under Ogata-binned arrivals, with mean-wealth difference -0.001 and paired 5/95 quantiles 0.002 and 0.049, reflecting a small number of negative tail paths. Absolute-Gamma is positive under both generators, with advantages 0.705 and 0.690 and paired 5/95 quantile ranges roughly $[-0.527, 2.651]$ and $[-0.524, 2.598]$. This check weakens any blanket claim that robust-Gamma dominates under every arrival generator, and it shows that the relative-slack result is generator-sensitive even though the absolute-Gamma result is similar across the two generators.

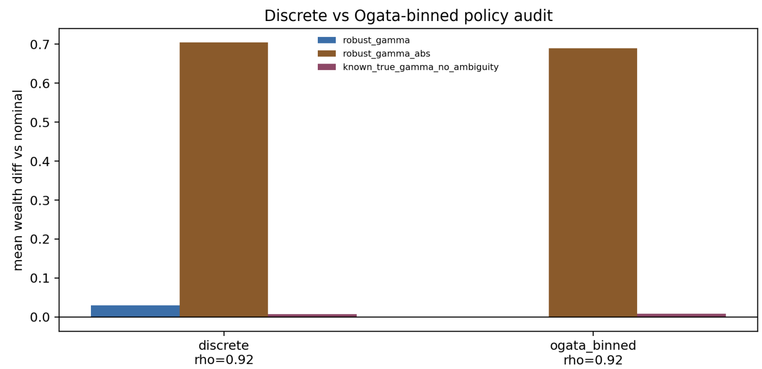

Figure 4: Focused policy check under fast discrete arrivals and Ogata-binned event arrivals for $\hat{\rho} = 0.92$.

6.5 Event-Queue Backtest

Finally, we run a reduced event-driven queue backtest on Ogata Hawkes event times. The evaluator holds the event path fixed across policies, updates quotes every $\Delta t = 0.02$, tracks a simple queue-ahead state, and records the side-time spent not quoting. For $\hat{\rho} = 0.92$ over 24 paths, robust-Gamma is nearly flat but positive relative to nominal, with mean-wealth difference 0.0125 and paired 5/95 quantiles $[0.004, 0.025]$, while absolute-Gamma and the liquidity guard each improve mean wealth by 0.342 with paired 5/95 quantiles $[0.115, 0.687]$. In this reduced run, fill rates are identical across policies at 0.943, the side-time no-quote fraction is 0.064, and the full no-quote time fraction is zero. This is not a production queue simulator, but it does provide an event-time execution stress test for the checked regime.

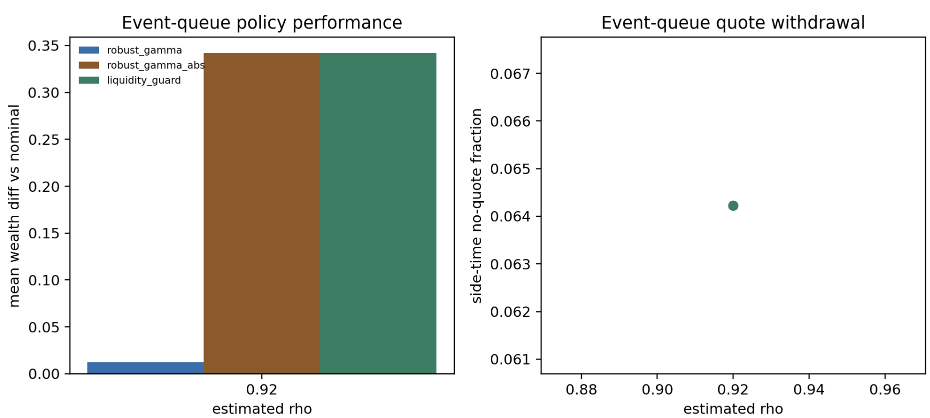

Figure 5: Reduced event-queue performance and shared side-time no-quote fraction.

6.6 Public LOBSTER Top-Of-Book Replay

The first replay asks whether the policy regimes observed in simulation also appear when replayed on public LOBSTER message/order-book streams. The replay uses up to the first 80,000 synchronized messages per ticker, posts active policies at the displayed L1 quotes subject to an inventory cap, depletes a simple queue-ahead proxy using observed execution sizes, and marks to the observed midprice. Under the calibrated side-aware spectral radii, all tested policies remain inside the L1 activity threshold, so the replay does not force an artificial separation: mean terminal wealth is 341.691, mean fills are 3071.8, and side-time no-quote share is 0.198 for each policy. Under the near-critical stress setting $\hat{\rho} = 0.97$, nominal and relative-slack robust-Gamma remain active on the same real event streams, while absolute-Gamma and the liquidity guard fully withdraw: their mean wealth difference versus nominal is -341.691, mean fills are zero, and side-time no-quote share is one. This should not be read as a production queue simulator. Its narrower point is that the quote-withdrawal transition appears in this public replay when the same policies are placed on observed L1 event paths and prices.

The next replay keeps the L1 restriction but makes quote placement visible. Each continuous policy offset is rounded into one of four displayed states: join the best quote, improve inside the spread, rest away from L1, or withdraw. Away quotes are not filled, because one-level data cannot identify hidden depth. Under calibrated side-aware spectral radii, tick rounding still makes policy PnL values coincide, but the state mix explains where the quotes would have gone: mean side-time is 0.024 joining, 0.428 improving, 0.448 away from L1, and 0.101 withdrawn. Under $\hat{\rho} = 0.97$, absolute-Gamma and the liquidity guard fully withdraw. This is a more informative public-data check than the binary replay, while still stopping short of production execution claims.

We then let away quotes interact with displayed depth by replaying the uniform level-10 LOBSTER panel. If a rounded quote rests inside the displayed ten-level ladder, the replay initializes queue-ahead from all better displayed levels plus a same-level queue fraction; observed executions and visible depth changes can then deplete that queue. Under calibrated side-aware spectral radii, policies still coincide after tick rounding, but the state mix is more informative: mean side-time is 0.018 joining L1, 0.428 improving, 0.408 resting at visible depth, and 0.146 withdrawn, with mean visible depth rank 2.431 and mean fills 1420.2. Under $\hat{\rho} = 0.97$, absolute-Gamma and the liquidity guard fully withdraw, with mean wealth difference -120.216 versus nominal. The replay remains conservative about hidden liquidity and exact priority, but the test is no longer L1-only.

The priority-aware replay adds explicit priority accounting within the public level-10 book. At each quote reset, the synthetic order is placed behind the full displayed size at the same price. Later same-price limit orders are tracked by order identifier as behind the synthetic quote, so cancellations or executions of those later orders cannot improve its queue position. Queued fills are credited only from residual same-price execution volume after displayed queue-ahead has been depleted; exact queue exhaustion with no residual volume is recorded but not filled, and partial fills are credited by residual lots.

This stricter rule shows how much the earlier displayed-depth replay depended on a heuristic same-level queue cap. Under calibrated side-aware spectral radii, mean fill events are 21.6 but mean filled lots are only 15.53, mean terminal wealth is -7.839, and mean side-time is 0.019 joining L1, 0.530 improving, 0.422 resting at visible depth, and 0.029 withdrawn. Under $\hat{\rho} = 0.97$, absolute-Gamma and liquidity-guard policies fully withdraw and avoid the nominal priority-replay loss, giving mean wealth difference $+5.964$ versus nominal. The regenerated artifact also records queue-position invariants: in the calibrated baseline, mean initial displayed queue ahead is 62.03 lots across 1810.4 visible quote resets, queued fills average 1.2, inside-spread improve fills average 20.4, partial-fill events average 7.6, the maximum zero-residual prevented-fill count is 1, and the maximum queue-violation count is zero. This is the most queue-position-aware replay in the package, but it still cannot observe hidden liquidity or recover order identifiers for anonymous queue ahead already present when the synthetic quote arrives.

As a depth supplement, we also replay the deepest public LOBSTER files available for the sample panel: level-50 AAPL and MSFT on their shorter one-hour public window. This supplement uses all 50 displayed levels in those files and records mean fill events 23.0, mean filled lots 16.82, mean queued fills 2.5, mean improve fills 20.5, mean partial-fill events 8.0, max visible depth rank 5, and maximum queue-violation count zero. Because only two tickers have public level-50 samples in the mirror, this is a supplement rather than the headline five-ticker panel.

We also run a priority-assumption sensitivity grid for the nominal policy. The grid varies the fraction of displayed same-price queue placed ahead of the synthetic order over $\{0, 0.5, 1\}$ and a queue-stress multiplier over $\{1, 1.5, 2\}$. The multiplier is applied to the public displayed queue-ahead proxy as a conservative stress test; it is not evidence that hidden depth has been observed. On the calibrated level-10 panel, the optimistic displayed priority setting $(0, 1)$ gives mean filled lots 16.53; the strict displayed-priority baseline $(1, 1)$ gives mean filled lots 15.53; and the most conservative tested queue-stress setting $(1, 2)$ gives mean filled lots 14.87. The narrow range is useful precisely because it is negative evidence: public displayed books rarely identify enough residual same-price execution volume to make hidden-priority assumptions strongly estimable. Across the sensitivity grid, the maximum queue-violation count remains zero.

Finally, we check whether the public messages support the queue analysis near the touch. We reconstruct the observable level-10 book from messages and compare it to the official LOBSTER snapshots. The reconstructor starts from the first official snapshot, tracks all post-start order IDs, and re-anchors every 100 messages so the check measures local reconstruction quality rather than all-day truncation drift. Across 399,995 processed events and 40,000 compared snapshots, panel mean top-of-book price match is 0.9997, full 10-level price match is 0.9376, top-of-book size MAE is 0.056 shares, and full-depth size MAE is 38.1 shares. These results are consistent with local near-touch priority analysis from public messages, while deeper queue-position claims should remain conservative.

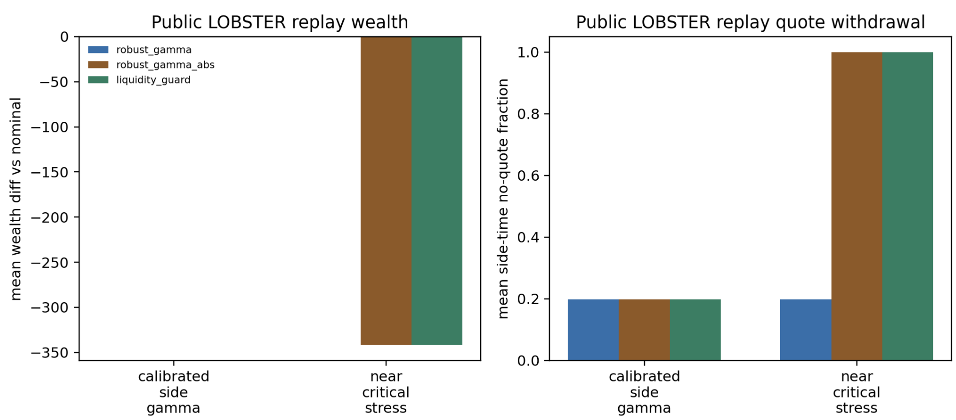

Figure 6: Public LOBSTER top-of-book replay on actual message/order-book streams. Calibrated policies coincide under the simple binary rule; in the near-critical stress case, absolute-Gamma ambiguity withdraws quotes on the same data paths.

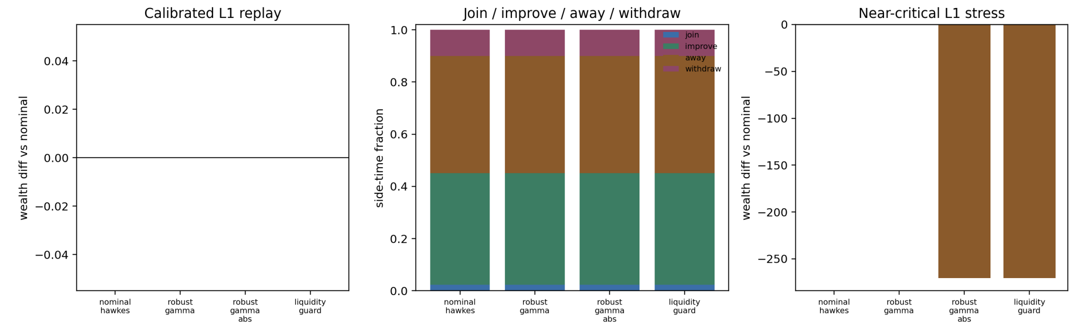

Figure 7: Offset-aware L1 quote replay. Tick rounding still makes calibrated policy wealth coincide, but the replay shows where quotes would be placed and when near-critical absolute-Gamma ambiguity withdraws them.

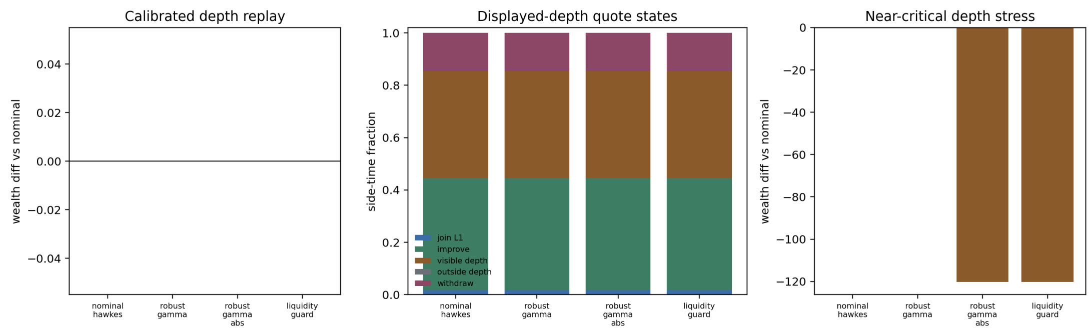

Figure 8: Displayed-depth level-10 LOBSTER quote replay. Away-from-L1 quotes can fill only when they rest inside the visible book and observed executions or depth changes deplete displayed queue ahead.

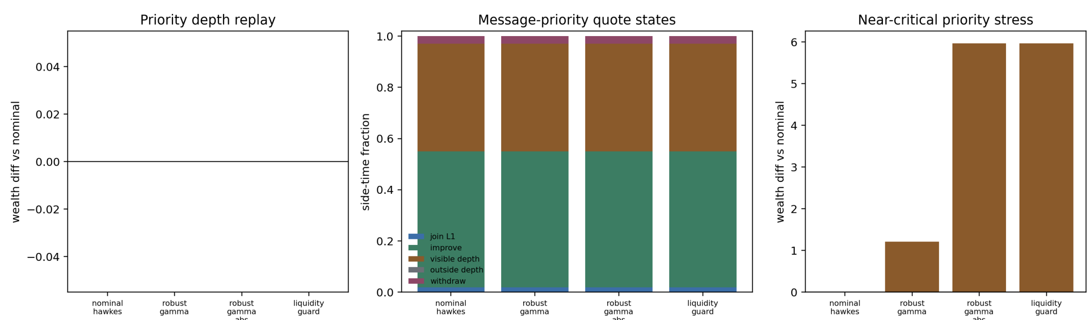

Figure 9: Residual-volume priority-aware level-10 LOBSTER quote replay. Same-price orders arriving after the synthetic quote are kept behind it; queued fills require residual execution volume after displayed queue-ahead depletion.

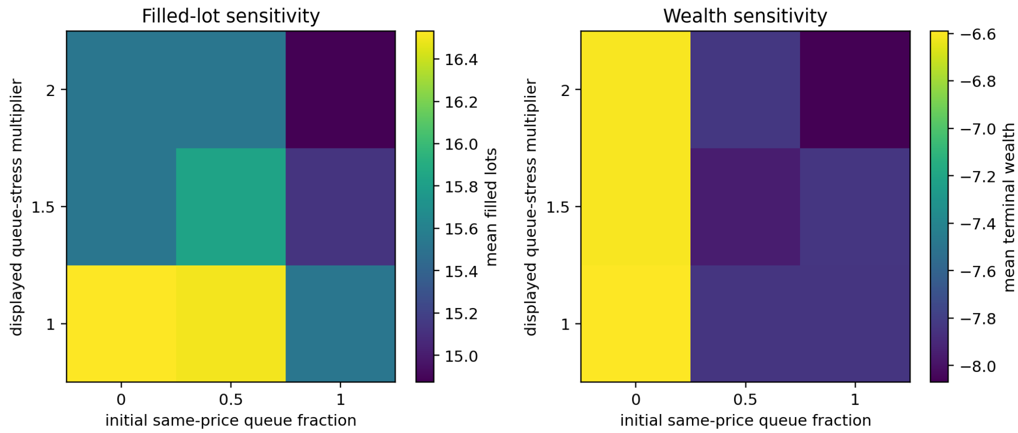

Figure 10: Priority-assumption sensitivity for the nominal level-10 replay. The grid varies same-price displayed priority and displayed queue stress; filled lots remain rare, showing the public-data limit of same-price priority identification.

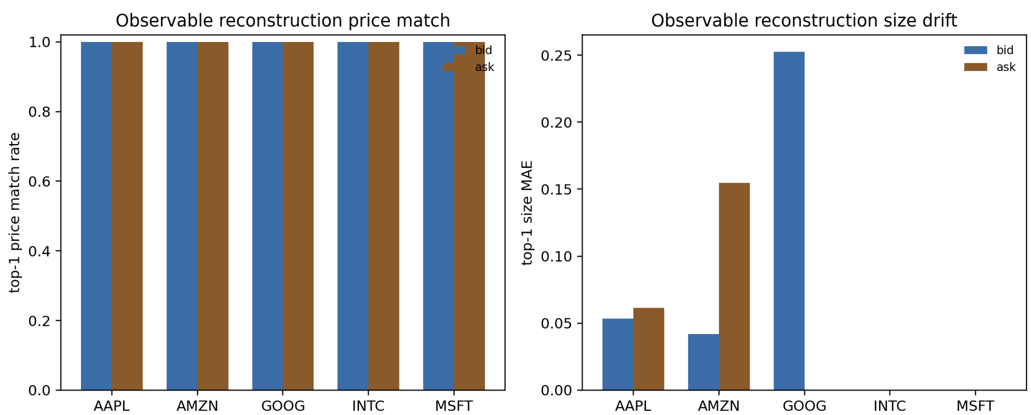

Figure 11: Observable order-book reconstruction check. Re-anchored message-level reconstruction matches top-of-book prices closely, while full-depth size drift remains visible.

6.7 Simulator Validation

The slow Ogata reference simulator makes the numerical issue visible: coarse time discretization can create spurious critical amplification. At $\rho = 0.90$, the relative mean-count bias falls from 3.07 at $\Delta t = 0.08$ to 0.03 at $\Delta t = 0.01$. At $\rho = 0.97$, even $\Delta t = 0.005$ leaves about 0.20 relative bias in the current short-horizon setting.

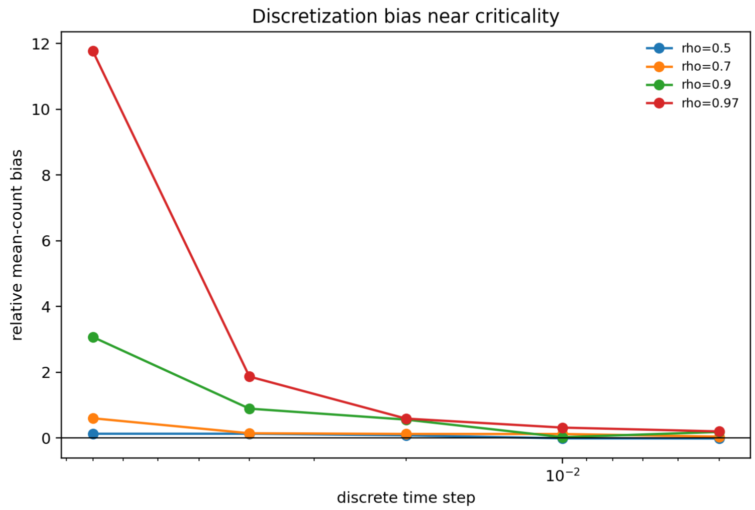

Figure 12: Time-discretization bias grows sharply near criticality.

6.8 Calibration Noise

A simple Fano-based criticality estimator points to the same practical concern: near-critical calibration is fragile. For $\rho = 0.97$, short horizons under-estimate criticality, while longer horizons can over-shoot because rare clusters dominate the count variance. This is why the empirical $\hat{\rho}$ estimates should be read with uncertainty, not as exact structural inputs.

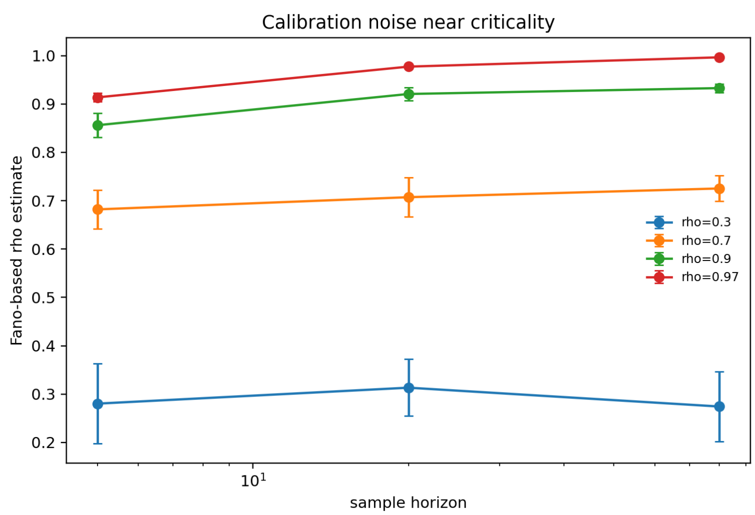

Figure 13: Fano-based criticality estimates across horizons.

6.9 Robust Dynamic Programming

The finite-scenario DP includes explicit bid-side and ask-side no-quote actions. In the reported reduced run, the raw DP action grid contains 4,044 two-sided quote states, 618 bid-only states, 618 ask-only states, and no full no-quote states. The first-step, zero-inventory rows remain two-sided through the tested grid, which reaches $\hat{\rho} = 0.92$. Averaged over the DP grid, no-quote rates are 22.3% under relative-slack ambiguity and 24.5% under absolute-Gamma ambiguity; full two-sided withdrawal rates are zero for both. This is still a reduced Bellman approximation, not a queue-calibrated quote-refusal simulator comparable one-for-one with [5].

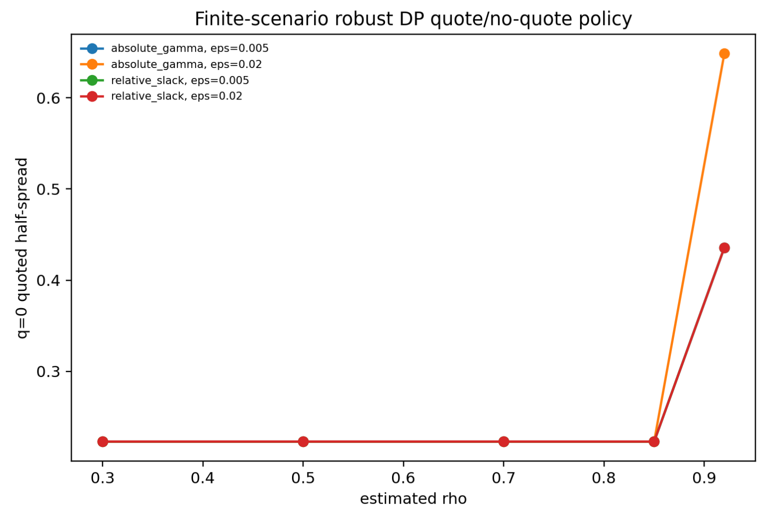

Figure 14: Reduced robust-DP half-spreads and no-quote switching as criticality increases.

We also run a quote-map sensitivity diagnostic linking the local optimizer theorem to the implemented policy map. For the smooth implemented half-spread, nominal and relative-Gamma robust policies recover critical exponent 3.000 for $d\delta/d\rho$, while absolute-Gamma robust policies approach exponent 4.000 in the uncapped interior region. Capped or no-quote points are marked as nonsmooth active-set cases rather than treated as classical derivatives.

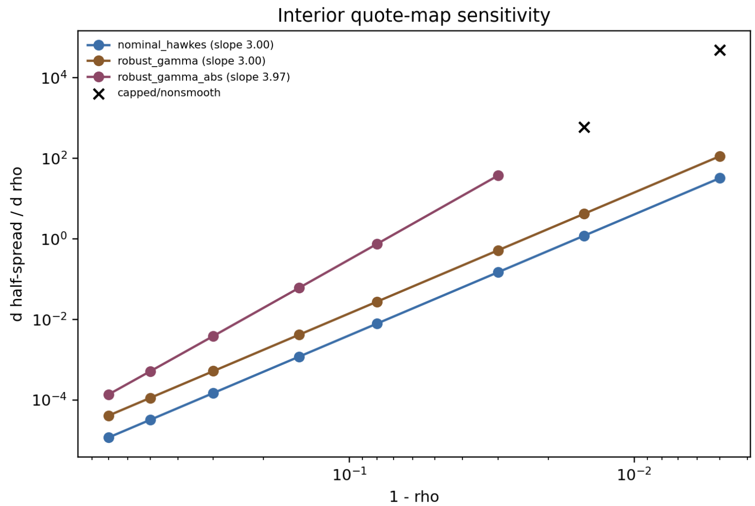

Figure 15: Interior quote-map sensitivity. Smooth relative-Gamma policies have third-order derivative blowup, while absolute-Gamma ambiguity adds one more critical derivative factor until spread caps bind.

6.10 Public Data Panels

The public datasets are not used to prove the structural ambiguity theorem, because the true Γ is unobserved. Instead, they document clustered order flow and overdispersed liquidity states in the data used for the diagnostics. Binance aggregate trades are fetched from the public archive at https://data.binance.vision/?prefix=data/spot/daily/aggTrades/.

Ticker	Rows	Events/sec	Event Fano	Mean spread
AAPL	118497	5.064	19.460	0.155
AMZN	57515	2.458	13.329	0.136
GOOG	49482	2.115	17.185	0.311
INTC	404986	17.307	220.650	0.013
MSFT	411409	17.582	196.485	0.013

Table 4: Public LOBSTER sample sanity panel.

Symbol	Agg. trades	Agg. trades/sec	Event Fano	ACF1
BTCUSDT	1364603	15.794	200.274	0.342
ETHUSDT	597417	6.915	29.730	0.241
SOLUSDT	218278	2.526	12.240	0.201

Table 5: Public Binance aggregate-trade event panel for 2024-01-15.

Symbol	Rows	Rel. spread bps	Return std	Depth Fano
BTC	10081	0.163	0.000687	33.989
ETH	10081	0.591	0.000910	513.703
SOL	10080	0.911	0.001110	1697.605

Table 6: Public crypto L2 depth sanity panel.

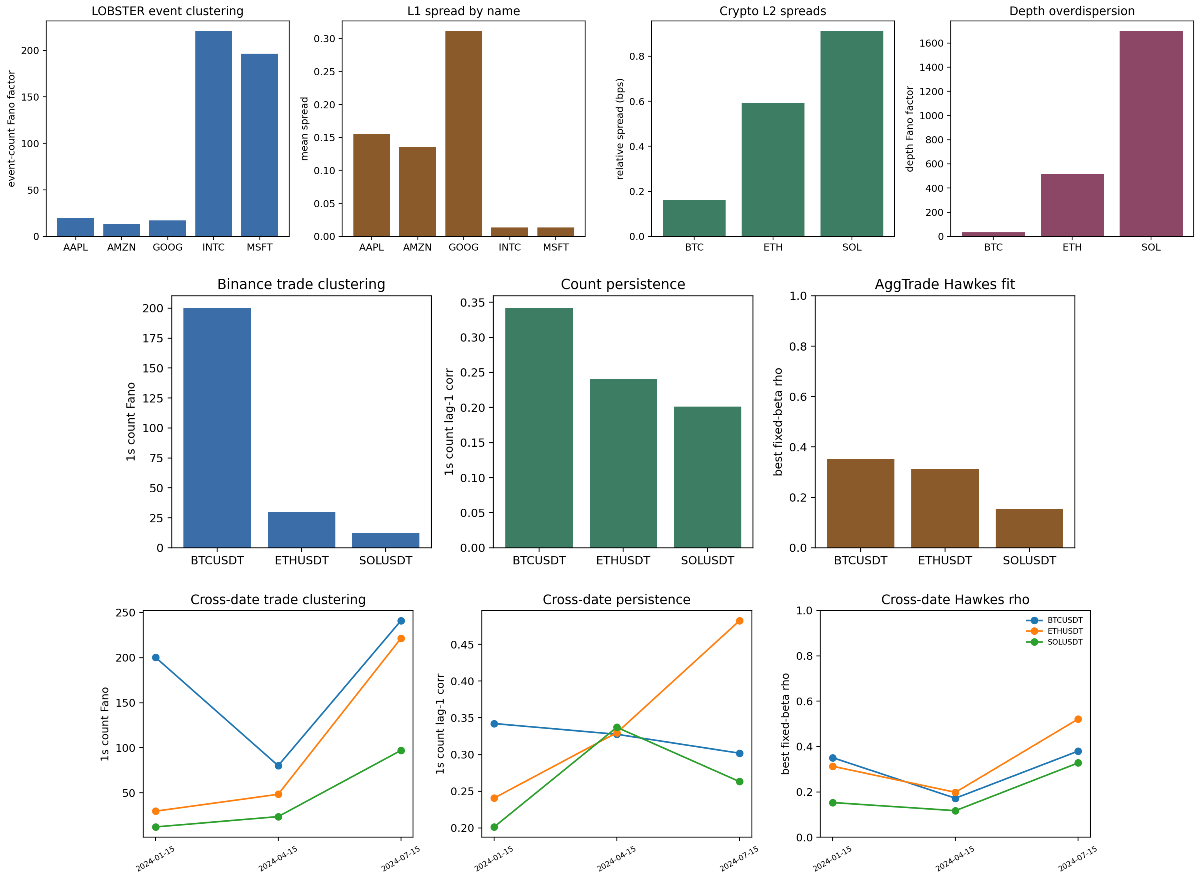

Figure 16: Public-data sanity figures: LOBSTER event clustering and spreads, crypto L2 spreads and depth overdispersion, Binance aggregate-trade clustering, fixed-beta Hawkes diagnostics, and cross-date robustness.

6.11 Corrected Hawkes Calibration Diagnostics

As a check before interpreting real-data Hawkes fits, we validate the corrected single-exponential MLE on Ogata-simulated paths with known parameters. The validation recovers moderate and near-critical branching ratios without beta-bound failures.

ρ true	β true	ρ MAE	β MAPE	Beta cap rate
0.30	2.0	0.043	0.201	0.000
0.70	4.0	0.019	0.026	0.000
0.90	8.0	0.016	0.039	0.000

Table 7: Synthetic Ogata validation for the corrected Hawkes MLE.

We also validate the fixed-beta marked multivariate estimator on three-type Ogata Hawkes paths with known branching matrices. For target spectral radii 0.35 and 0.65 at $\beta = 3$, the marked estimator has spectral-radius MAE 0.025 and 0.023, relative Frobenius branching-matrix error 0.234 and 0.133, and a success rate of 1.000. This check is important because the public-data marked calibrations below are only meaningful if the estimator can recover a known multitype Γ under controlled simulation.

ρ true	Mean events	ρ MAE	$\|\hat{\Gamma} - \Gamma\|_F / \|\Gamma\|_F$	Success
0.35	1209.8	0.025	0.234	1.000
0.65	2171.2	0.023	0.133	1.000

Table 8: Synthetic Ogata validation for the fixed-beta marked multivariate Hawkes estimator.

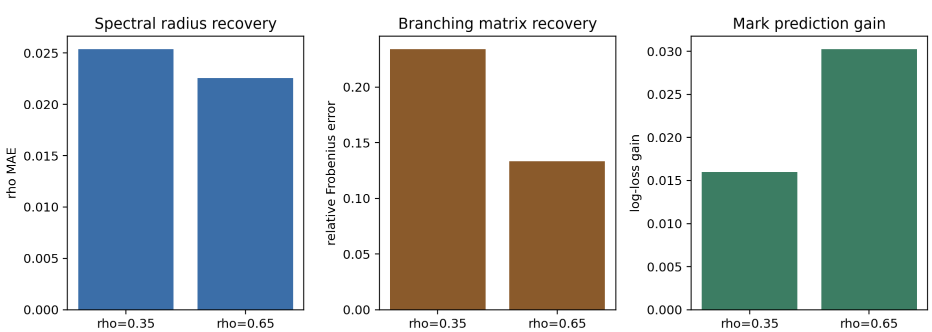

Figure 17: Marked multivariate Ogata validation: spectral-radius recovery, branching-matrix recovery, and conditional mark-prediction gain.

The Binance aggregate-trade panel provides a separate public check on crypto event clustering. On 2024-01-15, BTCUSDT, ETHUSDT, and SOLUSDT have 218k to 1.36m aggregate trades, one-second event-count Fano factors from 12.24 to 200.27, and lag-one count autocorrelations from 0.201 to 0.342. Fixed beta Hawkes fits over all aggregate trades, buy-aggressor trades, and sell-aggressor trades select $\beta = 100$ in the current grid. All-trade branching ratios are 0.351 for BTCUSDT, 0.313 for ETHUSDT, and 0.153 for SOLUSDT; sell-aggressor branching is higher for BTCUSDT and ETHUSDT, at 0.493 and 0.463. This is useful cross-asset evidence of clustered event flow, but aggregate trades do not include order placement, cancellation, or queue state, so the panel should not be read as an execution benchmark.

To avoid relying on one crypto day, we repeat the Binance check on 2024-01-15, 2024-04-15, and 2024-07-15. Across all-event fixed-beta fits, branching-ratio ranges are 0.172–0.380 for BTCUSDT, 0.198–0.521 for ETHUSDT, and 0.117–0.328 for SOLUSDT. The median all-event branching ratios remain 0.351, 0.313, and 0.153, respectively.

On raw LOBSTER message arrivals, the all-message single-exponential fit remains a misspecification diagnostic rather than a structural calibration target. The corrected all-message panel still places β at the upper bound for every ticker under the 100k-event cap, with fitted branching ratios between 0.64 and 0.76. Event-type splits and fixed-beta profiles show that branching estimates are sensitive to the assumed decay scale; timestamp aggregation sharply reduces the fitted decay, with median β falling from 148.4 on raw events to 23.4 at 1ms aggregation and 1.46 at 10ms aggregation. A fixed-beta two-scale diagnostic adds a multiscale check but does not remove the misspecification warning: among the tested pairs, every ticker/event-group selects $(\beta_{\text{slow}}, \beta_{\text{fast}}) = (1, 100)$, and median fast-excitation shares are 0.764 for all messages, 0.723 for limits, 0.660 for cancel/delete events, and 0.871 for executions. Execution events remain the hardest to fit, with median residual KS statistic about 0.415. This supports the methodological point that raw unmarked message flows mix multiple clocks and packet-level bursts, so, in this setting, robust market-making experiments should either use marked or multiscale calibration diagnostics, or report residual diagnostics [1, 17].

The package therefore also includes a grouped marked multivariate Hawkes fit on limit, cancel/delete, and execution event groups. With fixed exponential decay, AIC selects $\beta = 100$ for all five public LOBSTER tickers. The estimated branching matrices are stable, with spectral radii from 0.367 to 0.622, and the conditional mark log-loss improves over unconditional event-frequency prediction by 0.005 to 0.114 nats per event. The median branching matrix has the strongest source column from execution events, with median total outgoing branching about 0.923, and the largest median target row into limit intensity, about 0.897. Residual KS statistics still range from 0.050 to 0.217, so the fit should be read as a reproducible marked calibration diagnostic, not as a production-grade structural Γ estimate.

Finally, to align the empirical object with the bid/ask control model, we fit a six-mark side-aware Hawkes model using LOBSTER direction labels: limit, cancel/delete, and execution events on buy and sell sides. AIC again selects $\beta = 100$ for all five tickers. The full six-mark branching matrices remain stable, with spectral radii from 0.369 to 0.624, while conditional six-mark log-loss improves over unconditional mark frequencies by 0.101 to 0.292 nats per event. Median side-aggregated branching is strongest within side, about 1.354 from buy to buy and 1.403 from sell to sell, with smaller cross-side totals of about 0.492 from buy to sell and 0.414 from sell to buy. The aggregate row sums can exceed one because they sum many target/source entries; stability is assessed by the spectral radius of the full six-mark matrix.

We then condition the six-mark time-rescaling residuals on observable L1 state variables: event group, side, within-ticker size bucket, spread bucket, top-depth bucket, and order-book imbalance bucket. This is a misspecification check rather than another fitted model. Across the five public LOBSTER samples, wide-spread states have median residual mean 0.886 and median KS statistic 0.220, while tight-spread states have median residual mean 1.110 and median KS statistic 0.092. Large-size events have residual mean 0.828 versus 1.018 for small events. Execution marks show the largest conditional mark-prediction gain, with median log-loss improvement 0.384 nats per event. The result sharpens the empirical conclusion: side-aware excitation is informative, but the residuals still depend materially on size and book state.

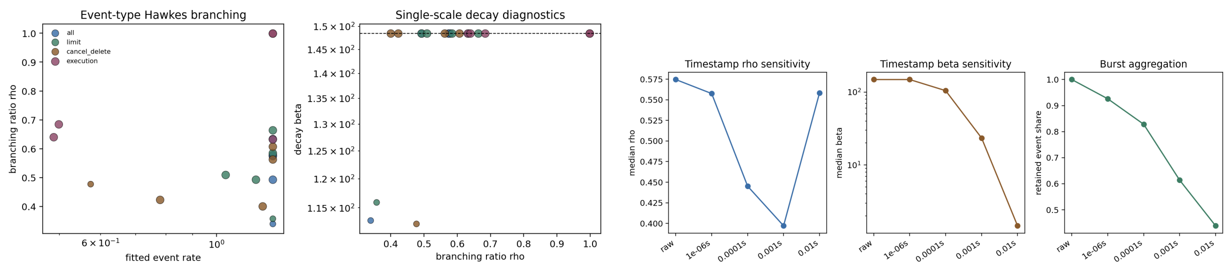

Figure 18: Corrected public-data Hawkes diagnostics: event-type branching/decay and timestamp-resolution sensitivity.

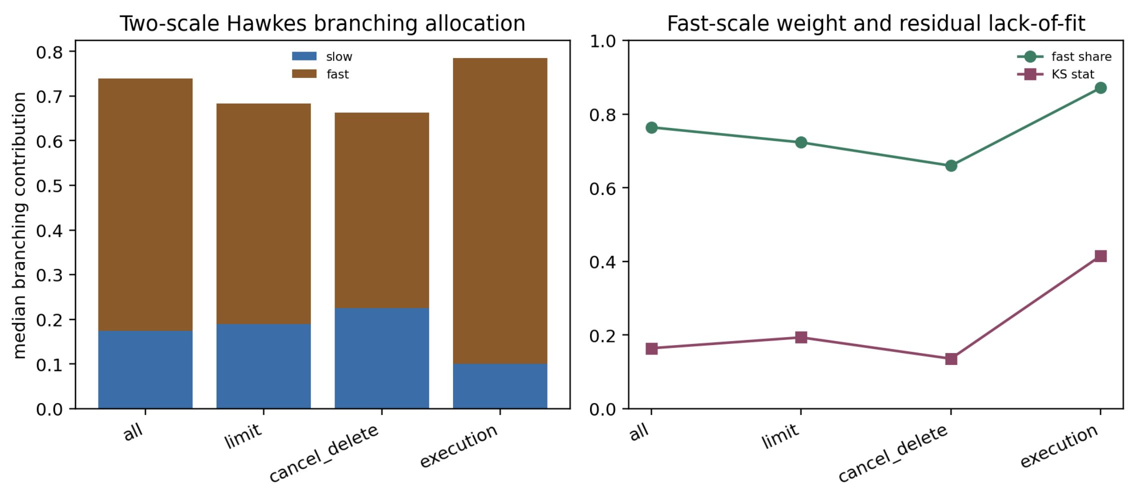

Figure 19: Fixed-beta two-scale public-data Hawkes diagnostics. The selected fits concentrate excitation on the fast component, especially for execution events, while residual tests still flag misspecification.

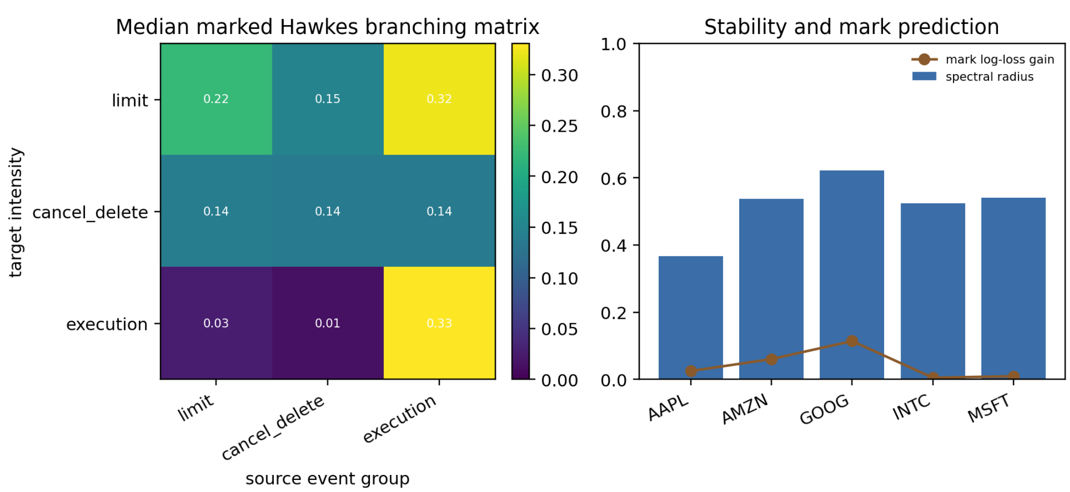

Figure 20: Grouped marked multivariate Hawkes diagnostics on LOBSTER limit, cancel/delete, and execution events. The fixed-beta fits estimate stable branching matrices and improve conditional mark prediction, but residual tests still point toward richer side, size, and multiscale calibration.

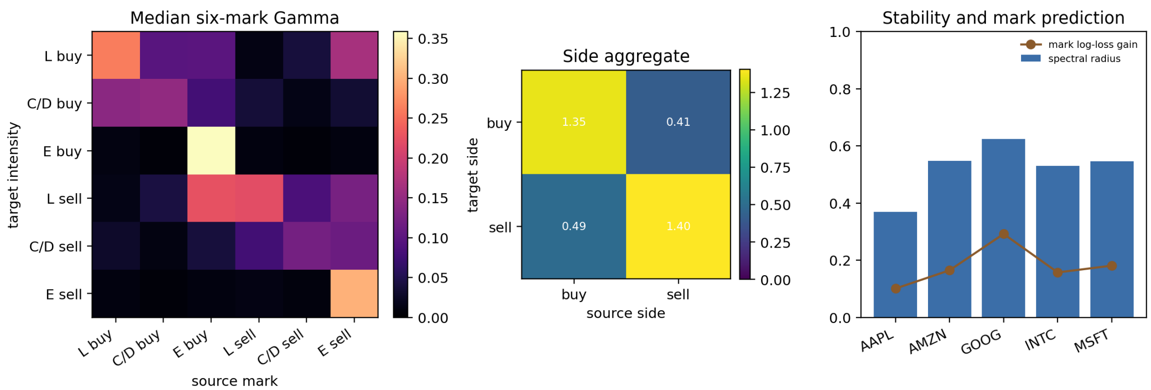

Figure 21: Side-aware marked multivariate Hawkes diagnostics using LOBSTER direction labels. The six-mark fits bring the empirical calibration object closer to bid/ask robust control, while remaining fixed-decay and message-side rather than queue-position or size-aware.

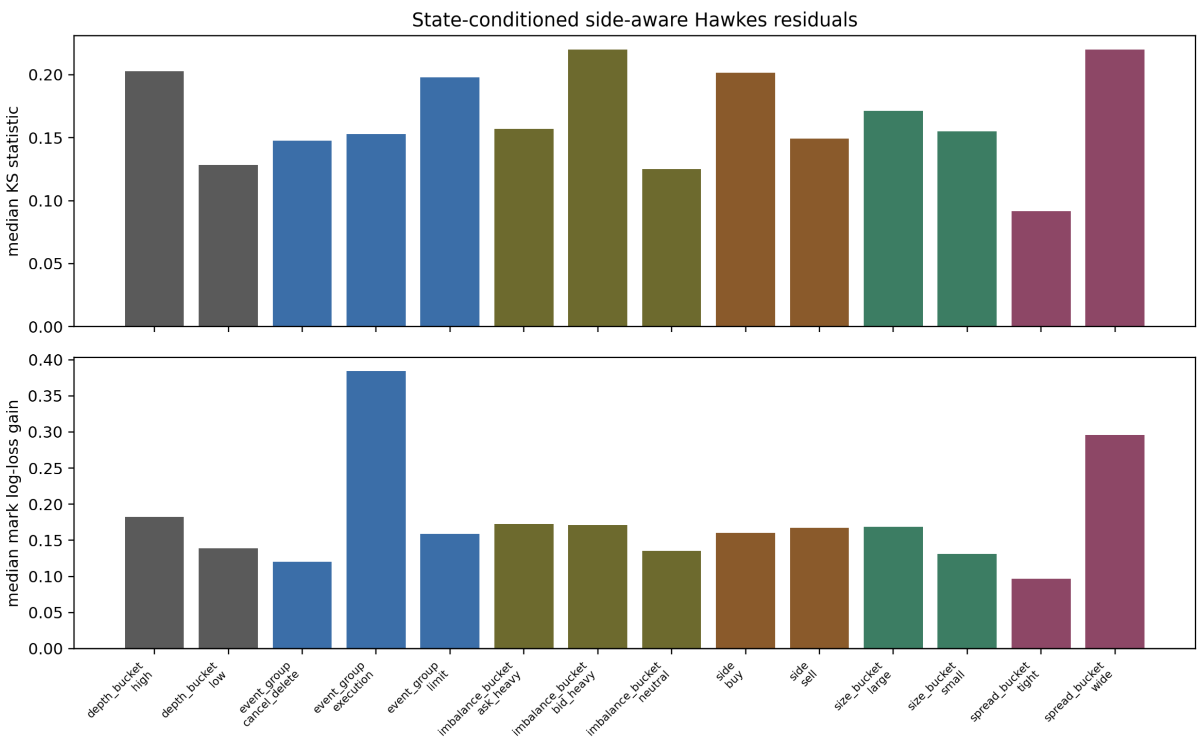

Figure 22: State-conditioned residual diagnostics for the side-aware six-mark Hawkes fit. Residual deviations and mark-prediction gains vary by spread, depth, imbalance, and event-size bucket, so the public-data calibration is best read as a diagnostic input rather than a complete structural law.

7 Limitations and Scope Conditions

This paper develops a conditional multitype HJBI control link for the stated exponential Hawkes model class. The argument combines compact truncated HJBI verification under compact-PDMP DPP/comparison hypotheses, untruncated Lyapunov stability with $p > 1$ tail control, pairwise weighted comparison and uniqueness in fixed-boundary W_v-weighted subclasses, local differentiability of smooth interior quote optimizers, and a consolidated argument assembling these pieces. It does not extend automatically to arbitrary state-dependent, learned, multiscale, or power-law Hawkes kernels without rechecking the stated Lyapunov and comparison assumptions. No-quote and active-adversary switches are treated as nonsmooth control boundaries, not as classical optimizer derivatives.

The top-of-book, L1, displayed-depth, residual-volume priority-aware level-10, deepest-public level-50, priority-sensitivity, and observable-reconstruction LOBSTER checks should be read as public-data execution checks rather than production execution claims, because hidden liquidity and anonymous initial queue priority remain unobserved. Binance aggregate trades should be read as event-clustering evidence rather than a queue-calibration source. These scope conditions delimit the paper's claim: the reported results concern structural Hawkes ambiguity, criticality amplification, and calibration fragility in the stated Hawkes market-making experiments.

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